

Project Management

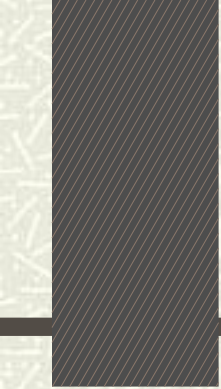
(MGMT 3192)

CHAPTER ONE

GENERAL INTRODUCTION

ABOUT PROJECT





What is project?
How can you define project
List the projects from your experience

1.1. Meaning and Definition of Project

What is project?

- # Project is in general a **building block** of an investment plan.
- # A project can be described as a combination of **human** and **material resources** pooled together in an organization **to achieve a specific objective.**

Definition of Project (Cont...)

According to USA Project Management Institute,

"Project is a one shot, time limited, goal directed, major undertaking, requiring the commitment of varied skills and resources"

It also describes a project as "a **combination of human and non-human resources pooled together** in a temporary organization to achieve a specific purpose."

Def. of Project (Cont...)

ISO-8402. (ISO document defines project as)

“A unique process consisting of a set of coordinated and controlled activities with start and finish dates, undertaken to achieve an objective conforming to specific requirement including constraints of time, cost and resources”

Def. Project (Cont...)

From the above all definitions we can learn that

- A project is an economic/development activity
- It requires commitment of scarce resources
- It brings some benefit from its accomplishment
- The benefit should **exceed** its cost
- Implementation of a project needs resources or **inputs**.
- Every project **converts** the given **inputs** into **outputs** through the process of implementation

1.2. Characteristics of Project

The special features of project that differentiate it from any other on-going activities, are

- # **Objective:** A project has a fixed set of objectives. Once the objectives are achieved the project ceases to exist.
- # **Life span:** A project can not continue endlessly. It has to come to an end at some time.
- # **Single entity:** A project is a single entity and is normally entrusted to one responsibility centre, while participants in the project are many.
- # **Life cycle:** A project has a life cycle reflected by growth, maturity and decay.

Characteristics of Project (cont...)

- # **Uniqueness:** No two projects are exactly similar even if the plants are exactly identical or are duplicated.
 - the location,
 - the infrastructure,
 - the agencies and
 - the people make each project unique.
- # **Team work:** Any project calls for team work-. The team members may be from different
 - discipline,
 - organization, and even country

Characteristics of Project (cont...)

- # **Change:** A project sees many changes throughout its life. While some of these changes may not have any major impact, there can be changes which will change the entire course of the project.
- # **Risk and Uncertainty:** Every project has a risk associated with it.

The degree of risk and uncertainty will depend on how a project has passed through the various stages of its life cycle.

Characteristics of Project (cont...)

■ **Unity in diversity:** A project is complex set of varieties. Varieties in terms of

- Technology
- Equipments and material
- People and machinery
- Work culture and ethics

But they remain interrelated.

High level of sub contracting : A high percentage of a work in a project is done through contractors. The more complex is a project, the more will be the extent of contracting.

1.3. Project Vs Plan/Program

- # Some people use the term '*project*' and '*program*' interchangeably.
- # In many cases the term project is used for what should be termed as program or work package.
- # However, there is a quite difference between the two.
- # A project normally originates from a plan which can be a national plan or corporate plan.

Program

Program in general is a group of related projects that are managed in a coordinated way to achieve certain objective.

A program is thus,

- # larger in scope,
- # not necessarily time bounded and
- # its objectives are broader

Program (Cont ...)

Example,

- # The national goal/plan: Poverty Eradication
- # Strategy: Increase productivity (in all sectors)
- # Development program: **Increase agricultural productivity**

This may result in a number of projects like,

- # Construction of dams (irrigation infrastructure)
- # Upgrading the skill of agricultural practices
- # Construction of training centers

Program (Cont ...)

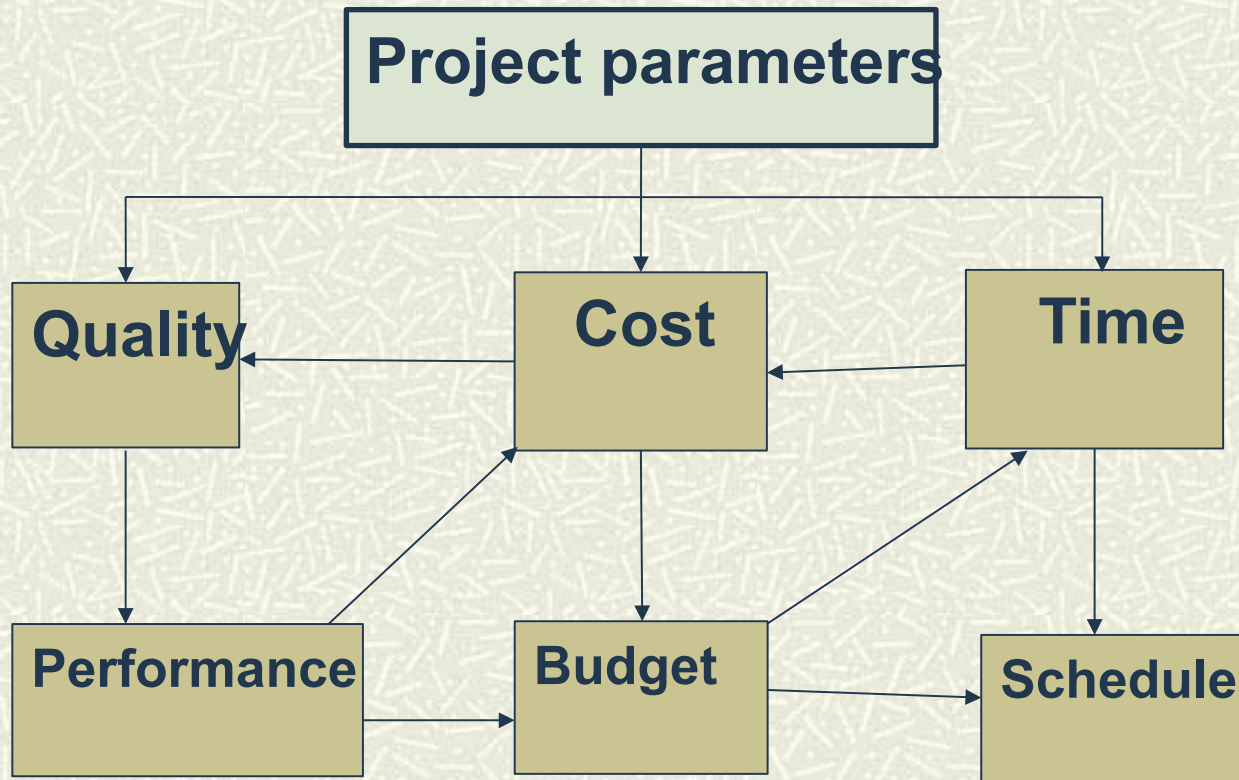
Health program may have a number of projects like,

- # Construction of hospitals
- # Training of health officers
- # Expansion of health centres.

1.4. Project Parameters

- ❖ During a project's life, management focuses on three basic parameters: **quality, cost, and time**.
- # Traditionally, it is said that a successfully managed project is the one that is completed at the specified level of quality, on or before the deadline, and within the budget.

Project Parameters (cont...)



Project Management

#What is project management?

Cont...

- ❖ **Project Management** is a set of principles, practices, and techniques applied to lead project teams and control project schedule, cost, and performance risks to result in delighted customers.
- **Cost, time and performance are the focus.**
- Project management includes **planning**, **organizing**, **directing**, and **controlling** project activities in addition to motivating what is usually the most expensive resource on the project - the people.

cont...

- # Project Management. Is therefore:-
- # The art of directing and coordinating human and material resources throughout the life of a project by using modern management techniques to achieve predetermined objectives of scope, cost, time, quality and participant satisfaction’.

Project management...Cont'd

- ❑ Every person, every organization and every nation is concerned with project management.
 - An individual builds a house. It is a project to him.
 - An organization sets up new factory. It is a project for the organization.
 - The government of a country builds high ways, dams, thermal power plants, hydropower plants, airports, etc. These are all projects that a country undertakes.

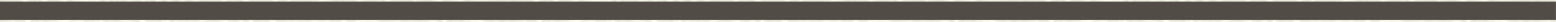
Typical Project Problems

1. Scope may not be clearly defined when commitment is made to a client.
2. There may not be enough resources allocated (people, money, materials, time, space, etc).
3. Conflict of interest between or among stakeholders (ops vs. engineers, sales vs. technical support, line vs. staff).
4. Commitment to unrealistic dates – the PM may be too optimistic about the completion date of the project.
5. There may be unclear roles and responsibilities.
6. Things may go wrong for some natural reasons.

Functions of the Project Managers

□ Project managers perform the following major functions/roles:

1. Plan work (scope, budget, schedule),
2. Obtain and manage resources,
3. Resolve conflicts and problems,
4. Motivate people
5. Communicate to the team, to the organization, and to the clients,
6. Set priorities,
7. Make decisions,
8. Control technical quality, budget, and schedule
9. Integrate multiple skills





CHAPTER TWO

THE PROJECT CYCLE

Meaning and definition of project life cycle

- ❑ Before any project is actually realized it goes through various planning phases
- ❑ The different stages through which project planning proceeds from inception to implementation are often called “the project cycle”.
- ❑ It is the project’s life cycle through which it advances from infancy to maturity.
- ❑ A project cycle is a sequence of events, which a project follows.

2.2 Project life cycle models

- ❑ Dividing project life cycle into phases helps in better management and control of a project.
- ❑ Like the definitions of a project, there are different project life cycle models which differ in their perspective, emphasis and level of detail.
- ❑ However, here we give more emphasis on the basic models. The Baum's cycle and other models such as UNIDO project cycle.

2.2.1. The Baum Cycle (World Bank Procedures)

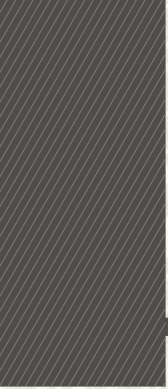


The first basic model of a project cycle is that of Baum (1970), which has been adopted by the world bank and initially recognized four main stages, and at a later stage (in 1978) the author has added an additional stage namely:

1. **Identification**
2. **Preparation**
3. **Appraisal and selection**
4. **Implementation**
5. **“Evaluation”**

Stage 1, Identification

- # The first stage in the cycle is to find potential projects.
- # Project identification amounts to finding project idea which contribute towards achieving specified business or development objectives.
- # The activity in this stage is coming up with project ideas



Stage 2, Preparation (pre – feasibility or feasibility studies)

Once projects have been identified, there begins a process of progressively more detailed preparation and analysis of project plans.

At this stage the project is being seriously considered as a definite investment action.

Project preparation (project formulation) covers the establishment of technical, economic and financial feasibility.

- # Decisions have to be made on the scope of the project, location and site, soil and hydrological requirements, project size (farm or factory size) etc.
- # Resource base investigations are undertaken and alternative forms of projects are explored.
- # Complete technical specifications of distinct proposals accompanied by full details of financial and economic costs and benefits are the outcome of the project preparation stage.

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- # Project design and formulation is an area in which local and international consultants are very active especially for big project that cover large areas and have big budgets.
 - # The project now exists as a set of *tangible proposals*.

Stage 3: Appraisal

- # After a project has been prepared, it is generally appropriate for a *critical review* or an independent appraisal to be conducted.
- # This provides an opportunity to re-examine every aspect of the project plan to assess whether the proposal is appropriate and sound before large sums are committed.
- # projects are appraised both in the field and at the desk level

✦ **Appraisals** should cover at least **seven aspects of a project**:

a) Technical – here the appraisals concentrate in verifying **whether what is proposed will work in the way suggested or not.**

b) Financial – the appraisals try to see **if the requirements for money needed by the project have been calculated properly, their sources are all identified, and reasonable plans for their repayment are made where necessary.**

c) Commercial – the way the **necessary inputs** for the project are conceived **to be supplied** is examined and the **arrangements for the disposal of the products** are **verified.**



d) Incentive – the appraisals see to it whether things are arranged in such a way that all those whose participation is required will find it in their interest to take part in the project.

e) Economic – the appraisal here tries to see the economic significance of the project

f) Managerial – this aspect of the appraisal examines if the capacity exists for operating the project and see if those responsible ones can operate it satisfactorily.

g) **Organizational** – the appraisal examines the project if it is organized internally and externally into units, contract policy institution, etc so as to allow the proposals to be carried out properly and to allow for change as the project develops.

- ❖ These issues are the subjects of specialized appraisal report. And on the basis of this report, *financial decisions are made – whether to go ahead with the project or not.*

Stage 4: Implementation

- # The objective of any effort in project planning and analysis clearly is to have a project that can be implemented to the benefit of the society.
- # Thus, implementation is perhaps the most important part of the project cycle.



In this stage,

- Funds are actually disbursed to get the projects started and keep running,
- A major priority during this stage is to ensure that the project is carried out in the way and within the period that was planned.
- It is during implementation that many of the real problems of projects are first identified. Therefore, to allow the management to become aware of the difficulties that might arise, recording, monitoring and progress reporting are important activities during the implementation stage.

Stage 5 Evaluation

- # The final phase in the project cycle is ***evaluation***.
- # Once a project has been implemented, it is often useful, to look back over what took place, to **compare actual progress with the plans**, and to judge whether the decisions and actions taken were responsible and useful.

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- # The extent to which the objectives of a project are being realized provides the primary criterion for an evaluation.
 - # The analyst looks systematically at the elements of success and failure in the project experience to learn how better to plan for the future.

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- # Evaluation should be undertaken when a project is terminated or is well into routine operation.
 - # Evaluation is not limited only to completed projects.
 - # It is important managerial tool in ongoing projects.
 - # And formalized evaluation may take place at several times in the life of a project.

- # Different people may do **evaluation**.
- # *Project management* will be continuously evaluating the project
- # *The sponsoring agency*, perhaps the operating ministry, the planning agency or an external assistance agency – may undertake evaluation.
- # In large and innovative projects, the project's administrative structure may provide a separate *evaluation unit*

UNIDO – Project Cycle

UNIDO has established a project cycle comprising three distinct phases):

1. THE PRE INVESTMENT PHASE
2. INVESTMENT PHASE AND
3. OPERATIONAL PHASE.

- Each of these *three phases* is divided into stages, some of which constitute important *consultancy, engineering and industrial* activities.

1. The *pre- investment phase* comprises:

- ❑ The pre-investment phase comprises several stages:
 - i. Project Identification
 - A. *Opportunity studies:*
 - B. *Preliminary Screening*
 - C. *Pre-feasibility Studies*
 - ii. Project preparation (feasibility studies)
 - iii. Project appraisal and investment decisions (**appraisal report**).

a) *Opportunity studies:*

- # The identification of investment opportunities is the starting – point in a series of investment – related activities.

Sources of Project Idea

1. At the macro level project ideas come from

- # National, sectorial, or regional plans and strategies
- # Unusual events such as drought, flood, earthquake etc
- # Multi or bilateral agreement of countries

2. At micro level project idea emanates from:

- # The identification of **unsatisfied demand or need** of the society
- # The existence of **unused or underutilized natural or human resources**
- # The initiative of private or public enterprise in response to the **incentives provided by the government**



b) ***Preliminary Screening:*** Once some project ideas have been put forward, the first step is **to select one or more of them as potentially viable.**

- # This calls for a **quick preliminary screening by experienced professionals** who could also modify some of the proposals.
- # At this stage **the screening criteria are vague and rough**, becoming specific and refined as project planning advances.

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- # During preliminary selection the analyst should eliminate project proposals that are
- Technically unsound and risky;
 - Have no market for the output;
 - Have inadequate supply of inputs;
 - Very costly in relation to benefits;
 - overambitious sales and profitability;

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- # It Depends on the experience and impartiality of the professionals applying them.
 - # It is, however, necessary to conduct this screening, even with indistinct criteria, in order to reduce to a manageable number the project alternatives to which more work and time will be devoted.



c) *Pre-feasibility Studies:* Following the preliminary screening, promising project options should be investigated in a systematic manner.

This requires the preparation of brief reports that indicate in sufficient detail the project versions that are still promising and suggest which ones should be eliminated

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- # *A pre – feasibility study* should be viewed as an intermediate stage between a project opportunity study and a detailed feasibility study
 - # The difference being in the degree of detail of the information obtained and the intensity with which project alternative are discussed.



✦ **Content of the Pre-feasibility Study:** To enable the relevant authorities to decide on the merits of various project options, the pre-feasibility study should, although briefly, discuss:

- i. The structure and objectives of the project;
- ii. The nature and size of the demand for the output or the needs that would satisfy, together with the foreseen beneficiary groups;
- iii. The availability of the most important materials and human inputs;
- iv. Basic alternative technologies available and their merits and weaknesses;

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- v. Approximate investment and operation costs as well as expected revenues and other benefits;
 - vi. Rough estimates of financial and economic returns;
 - vii. Any major factor that is likely to have an important effect on the project

ii. *Project Preparation – Feasibility Studies*

- ✦ If the pre-feasibility study indicates that the project is promising and further work is justified, the project enters the stage of preparation or feasibility study
- ✦ A feasibility study should **provide all data necessary for an investment decision.**
- ✦ **The commercial, technical, financial, economic and environment prerequisites for an investment project should therefore be defined and critically reviewed**

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- # The analysis of the project's **marketing, technical, financial, economic and institutional aspects** should be comprehensive enough to allow the policy makers to decide on the future of the project with confidence.
 - # Project preparation takes the form of a feasibility study **conducted by the agency sponsoring the project or by consultants.**

iii. *Appraisal and Investment Decision*

- # The project proposal in the form of a feasibility study is submitted to the **investment decision makers for a broad and impartial appraisal.**
- # **Appraisal** is the **comprehensive and systematic assessment of all aspects of the proposed project.**
- # After appraising the project carefully, **appraisers will decide whether it will be implemented or not, with or without minor modifications.**

2. THE INVESTMENT PHASE

- # The next stage in the project is the **actual implementation of the project**, followed by operation
- # While in earlier stages of **project planning there was more thinking and less action**, in this stage the combination switches in favour of the latter: **more action and less thinking is needed.**



The phase is divided into the following stages:

- 1) **Establishing the legal, financial and organizational basis** for the implementation of the project.
- 2) **Technology acquisition and transfer,**
- 3) **Detailed engineering design and contracting, including tendering, evaluation of bids and negotiations.**



Detailed engineering design will include site preparation final selection of technology construction planning and time scheduling, Negotiations are concerned with legal obligations arising from the acquisition of technology, construction of buildings, purchase and installation of machinery

4) Acquisition of land for construction work and installation.

5) Pre-production marketing,



6) Recruitment and training of personnel.

7) Plant commissioning and start-up. It is usually a brief but technically critical span in project implementation.

- ❖ It links the preceding construction phase with the operational (production) phase.

- ❖ This stage covers the signing of contracts between the investor on the one hand, and the financing institutions, consultants, architects and supplies of raw materials and required inputs on the other.

3) OPERATIONAL PHASE

- ✦ This is the **production phase** that commences after commissioning and **start-up of production**, replacement and rehabilitation, and expansion and innovation.
- ✦ The resultant **challenges** of this phase are viewed from the **short-term perspective** and **long-term perspective**.
- ✦ In the **shot-term challenges** may arise with regard to application of **production techniques, operation of equipment, inadequate labour productivity** due to a lack of qualified staff and labour etc.
- ✦ The **long-term** view relate to **the chosen strategies** and the associated **production and marketing costs** as well as sales revenues.

CHAPTER THREE

PROJECT IDENTIFICATION AND SCREENING



■ Project Idea- Meaning

- #Project idea identification is the first step towards establishing a successful venture.
- #This stage is about finding potential projects that will result in positive net present value or that provide benefits for the society.
- #Projects usually start as new ideas which are carefully examined and if found feasible and desirable are translated in to projects. Such initiation of project ideas is called project conception.

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- # **Project conception** is forming or developing ideas regarding a required intervention in a specific area to address a problem or take advantage of opportunities.
 - # **Project identification** is concerned with collection, compilation and analysis of economic data for the purpose of locating possible opportunities for investment and with development of such opportunities.



Sources of Project Ideas:

The sources of project ideas are mainly classified into **two types**. The classifications are as follows.

1. **Macro Source of Project Idea**

2. **Micro Sources of Project Idea**

1. At the macro level project ideas come from

- # National, sectorial, or regional plans and strategies**
- # Unusual events such as drought, flood, earthquake etc**
- # Multi or bilateral agreement of countries**

2. At micro level project idea emanates from:

- # The identification of **unsatisfied demand or need** of the society
- # The existence of **unused or underutilized natural or human resources**
- # The initiative of private or public enterprise in response to the **incentives provided by the government**

3.4. Opportunity Studies and Preliminary Screening

- #By using the above two sources of project idea, *a long list of project ideas can be developed.*
- #However, *those projects that are not promising should be eliminated* through a preliminary screening process.



✦ This preliminary screening weighs a project in terms of the following variables:

A. **Compatibility with the Promoter:** The idea should be compatible with the interest, personality and resources of the entrepreneur.

❖ A real opportunity should fit the personality of the promoter-abilities, training and priorities; accessible and offers a prospect of rapid growth and high return on invested capital.

B. Compatible with Governmental Priorities : The project idea should be within the government's regulatory framework. In addition, it should be in line with the country's goals to ensure its sustainability.



C. Availability of Inputs: The resources and inputs required for the project must be reasonably assured. These inputs include: **capital requirements of the project, technical know-how, raw materials, power and foreign exchange.**

D. Adequacy of Markets: The size of the market for the goods or services produced by the project must offer adequate sales volume. In this regard, **both the current and predicted demand levels should be assessed.**



In general assessment of the market adequacy should consider:

- a. Total present domestic market
- b. Competitors and their market share
- c. Export markets
- d. Sales and distribution system
- e. Projected increase in consumption
- f. Barriers to entry
- g. Economic, social and demographic trends
- h. Patent protection

E. *Investment costs:* - Costs of raw materials transportation costs, distribution costs, labour costs, production costs and investment costs are usually considered. If the above costs are very high sustainability of the project is questionable.

Demand and supply factors - This involves the projection of both short term and long term requirements of the project's output and examining the implications of these on the project's capacity. Many projects fail because they have started with a

F. Social and environmental considerations: - the project should also conform to the social considerations of the locality in which it is implemented.

In addition; especially in industrial projects, due consideration should be given to the environmental effects of it on the surrounding.



G. Reasonableness of Costs: The cost of the proposed project should be commendable considering the benefits expected from it.

✦ The following costs are considered in this regard:

- Cost of material inputs and labour
- Factory overheads
- Administrative expenses

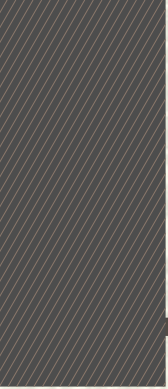


H. Acceptability of Risk Level: The desirability of a project is critically dependent on the risk characterizing it.

- The following factors are used in assessing the risk level of a project:
- Vulnerability of business cycle
 - Technological changes
 - Competition from substitutes and imports
 - Government control over price and distributions

3.5 Pre-feasibility Studies

- It is an intermediate stage between the project identification and a detailed feasibility study. The difference between the feasibility study and the pre-feasibility study is in the extent of detail of the information obtained.
- The structure of the pre-feasibility study should be the same as the structure of the feasibility study.



In particular, the review should cover the various alternatives identified in terms of:

- Project or corporate strategies and scope of the project
- Market and marketing concept
- Raw materials and factory supplies
- Location, site and environment
- Engineering and technology
- Organization and overhead costs
- Human resource
- Project implementation schedule and budgeting



In general, the pre-feasibility study involves subjective judgment of the project in terms of:

- A. Availability of an adequate market:* - judgments relating to the
- ✓ Number of potential customers,
 - ✓ Needs of the customers,
 - ✓ Strength of the competitors,
 - ✓ Availability and access to sales and distribution network and export possibilities.



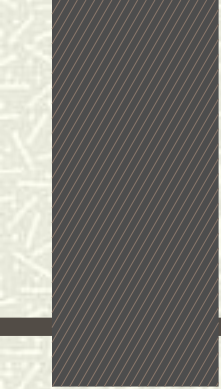
B. Project growth potential: - assessment of indicators on projected increase:

- ✓ In the number of customers,
- ✓ Increase in the rate of acceptance of the products,
- ✓ The general economic, social and political trend which could affect the growth potential of the project.



C. Investment costs: - Costs of raw materials transportation costs, distribution costs, labour costs, production costs and investment costs are usually considered.

If the above costs are very high sustainability of the project is questionable.



D. Demand and supply factors:

E. Social and environmental considerations: - the project should also conform to the social considerations of the locality in which it is implemented.

- In addition; especially in industrial projects, due consideration should be given to the environmental effects of it on the surrounding.



CHAPTER FOUR

PROJECT PREPARATION (FEASIBILITY STUDIES)



CHAPTER FOUR

4. Project Preparation

4.1. Market and Demand Analysis:

- ❑ The first step in project analysis is to estimate the potential size of the market for the product (or service to be offered) proposed to be manufactured and get an idea about the market share that is likely to be captured.

- ❑ The key steps involved in market and demand analysis are organized into seven sections as follows:
 - a. Situational analysis and specification of objectives
 - b. Collection of secondary information
 - c. Conducting market survey
 - c. Characterization of the market
 - c. Demand forecasting
 - c. Uncertainties in demand forecasting
 - c. Market planning.

A. Situational Analysis and Specification of Objectives

- ✦ A situation analysis provides one with the current status of the key aspects of the market and its participants.
- ✦ It include customers, competitors, and middlemen.
- ✦ It analyses the characteristics of this aspects such as customer's preferences and purchasing power, strategies and actions of competitors, and practices of middlemen.

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- # To illustrate suppose Sony Corporation developed technology to produce superior plasma television set and the management wants to know where and how to market the TVs. The objectives of the market and demand analysis may be to answer the following questions:
 - # Who are the buyers of the TVs?
 - # What is the current demand for the TVs?
 - # How is the demand distributed geographically and seasonally?
 - # What prices will the customers be willing to pay for the superior TVs?

B. Collection of Secondary Information

- ✚ Information may be obtained from secondary and /or primary sources. Secondary information is information that has been gathered in some other context and is already available.

General sources of Secondary information:

- National level Report prepared by the Government
 - Economic Survey
 - Annual Reports by the Ministry
 - Publication of Advertising agencies
- ✚ Secondary information should be used with great caution. Its reliability, accuracy, and relevance for the purpose on hand should be determined before use.

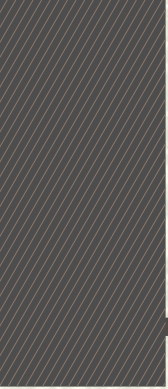
C. Conduct of Market Survey

- ✚ Secondary information, though useful often, does not provide a comprehensive basis for market and demand analysis.
- ✚ It needs to be supplemented with **primary information gathered through a market survey, specific to the project being appraised.**
- ✚ The survey could be **a census**; where the entire population is covered or **a sample survey**; only a proportion of the population is covered.
- ✚ Census studies are prohibitively costly and infeasible because proper sampling can produce the same or even more accurate results than census studies.



Information sought in a market survey may cover one or more of the following areas:

- Total demand and rate of demand growth
- Demand in different segments of the market
- Income and price elasticity of demand
- Motives for buying
- Purchasing plans and intentions
- Satisfaction with existing product
- Unsatisfied needs
- Attitude towards various products
- Social economic characteristics of buyers
- Customer preferences etc



Steps in a sample survey:

1. Select the sampling method and sample size.
2. Develop the questionnaire
3. Recruit and train the field investigators
4. Obtain information as per questionnaire from the sample respondents
5. Scrutinize the information gathered
6. Analyze and interpret the information.

- 
- The results of a market survey depend on the following:
 - representativeness of the sample,
 - precision and adequacy of the questions,
 - comprehension of the questions by the respondents,
 - honesty of the respondents in answering the questions,
 - integrity of the investigators and appropriateness of data analysis methods.

D. Characterization of Market

- ❑ Based on the information gathered from secondary sources and through the market survey, the market for the product/service may be described in terms of the following:-
 - Effective demand in the past and present
 - Methods of distribution and sales promotion
 - Price
 - Consumer
 - Supply and competition

E. Demand Forecasting

- # After gathering information about the various aspects of the market and demand from primary and secondary sources, attempt may be made to estimate future demand.
- # A wide range of forecasting method is available to the market analyst. These methods can be grouped in two categories

I) Qualitative Methods: This method essentially rely on the judgments of experts

A) Jury of executive opinion method: under this method opinions are sought from a group of managers on the expected future sales they are then translated into sales estimates

B) Delphi method: opinions are sought from a group of experts who don't know the identity of each other, any divergent opinions are then mailed back to back for further opinion until a consensus is obtained.

II) Quantitative Method: This method essentially rely on Quantitative data

- # *Trend projection method:*
- # *Time Series Projection Methods*
- # *Exponential Smoothing Method:*
- # *Moving Average Method:*
- # *Casual methods*
- # *Chain ratio method:*
- # *Consumption level method:*

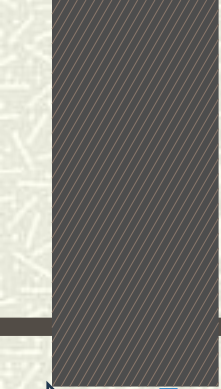
f. Uncertainties in Demand Forecasting: Demand forecasts are subject to error and uncertainty which arise from three principal sources:

- Data about past and present market
- Methods of forecasting
- Environmental change

g. Market Planning. Prepare a marketing plan for the new product.

4.2 Material Inputs and Supplies Study

- ✦ Different materials and other inputs required for operating the project should be identified and **their availability**, **supply** and **method of estimating operating costs** should be analyzed



➤ In this part of the feasibility study the following can be included:-

- i. Identification of the type of material inputs and supplies to be used in the project.
- *Raw material* (processed and/or semi processed) are those materials which can be agricultural products, mineral products, livestock and forest products, marine products)
 - *Processed Industrial material and components* (metal, semi processed materials, components and sub components)
 - *Auxiliary materials and factory supplies* (chemicals, additives, packaging materials, paints, oil, greasing, cleaning material, etc)
 - *Utilities*: a broad assessment of utilities (power, water,)

- ii. All requirements of materials and supplies should be identified and specified in the study considering all socio, economic, commercial, financial, and technical factors.
- ii. The source of materials availability, and price of inputs are to be analyzed.
- ii. location of the available resources, area of supply, access to transport, transport costs and alternate usage of such materials need to be collected.

v. Costs of raw materials and supplies:-

- ✓The costs of materials and other supplies have to be analyzed in detail to determine project economies.
- ✓Estimating annual operating costs for materials and supplies are to be made explaining the price mechanisms and key factors affecting prices.
- ✓Cost estimates may be expressed either as the cost per unit produced or in terms of a certain production level to conduct sensitivity analysis.

4.3. Location, Site and Environmental Impact Assessment

4.3.1. Location

- Location analysis has to identify locations suitable for the industrial project under consideration.
- **Location and site are often used synonymously but must be distinguished.** The choice of location should be made from a fairly wide geographical area, within which several alternatives sites can be considered.

- 
- Traditional approach to industrial location focused, on the proximity of raw materials and market place, mainly with the intention of minimizing transport costs.
 - However, the modern view requires consideration of not only commercial, technical and financial factors, but also of the social and environmental impact a project might have.



Generally, the choice of location is influenced by a variety of considerations: proximity to raw materials and markets, availability of infrastructure, labor situation, governmental policies, and other factors.

A) Proximity to Raw Materials and Markets

- ✚ Proximity to the sources of raw materials and nearness to the market for the final products are an important considerations for location. In light of a basic location model, optimal location is one where the total cost (raw material transportation cost plus production cost plus distribution cost for the final product) is minimized.

✦ **Practically, it means that:**

- A resource – based project like a cement plant or a steel mill should be located close to the source of the basic material (for example, limestone in the case of a cement plant and iron ore in the case of a steel plant;
- A project based on imported material may be located near a port
- A project manufacturing a perishable product should be close to the center of consumption.

- 
- # **Availability of Infrastructure:** In a feasibility study, availability of power, transportation, water, and communications should be carefully assessed before a location decision is made.
 - # **Labor Situation:** In project where there is labor-intensive, the labor situation in a particular location becomes important. The key considerations in evaluating the labor situation are:
 - Availability of labor, skilled, semi – skilled and unskilled
 - Existing labor rates
 - Labor productivity
 - Labor legislation

Governmental Policies

- # Policies and regulations of a government have a considerable influence on location. *In most of the cases of public sector projects, location is directly decided by the government.*
- # *In the case of private sector projects, location is influenced by certain governmental restrictions and inducements.* Most often the government may forbid the setting up of industrial projects in certain areas which suffer from urban congestion.

Other factors

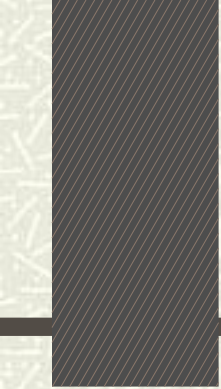
- ❑ **Climatic conditions** (Information should be collected on temperature, rainfall, flooding time) and **Geophysical aspects** (soil, water levels, and a number of special site hazard such as earthquakes and susceptibility to flooding)
- ❑ **General Living Conditions:** the general living conditions, such as the cost of living , housing situation, safety, and facilities for education, health care, transportation and recreation need to be assessed carefully.

Site Selection

- ✦ Once the location is decided upon, a specific project site and, if available, site alternative should be defined in the feasibility study.
- ✦ A project site selection may depend on particular site condition, which should be identified and described in the feasibility study.
- ✦ Two or three alternative site must be considered and evaluated with respect to:
 - Cost of land
 - Cost of site preparation and development

Ecological impact of the project, environmental impact assessment (EIA)

- ❑ The site environment impact analysis will cover the impact of the project and the alternatives (in terms of size, technology etc) on the surrounding area, including its population, flora and fauna.
- ❑ This analysis should be integrative and interdisciplinary, assessing the overall impact which taking into account the synergetic effects of inter-linked systems.

- 
- ✦ It is part of the project planning process and an integral part of feasibility analysis.
 - ✦ Environmental benefits or costs of a project are usually externalities or side effects that affect the society in whole or in part.

Objectives of Environmental Impact Assessment

- ✦ The general objective of environmental impact assessment in, project analysis is to ensure that development projects are environmental sound.



The specific objectives of environmental impact assessment are as follows:

- To promote a comprehensive, interdisciplinary investigation of environmental consequences of the project
- To identify critical environmental problems requiring further investigation
- To develop an understanding of the scope and magnitude of incremental environmental impacts
- To identify measures for mitigation of adverse environmental impacts and for possible enhancement of beneficial impacts.

4.4 Production Program and Plant Capacity

4.4.1 *Production program*

- ✦ Production program is a plan consist of the whole range of project activities and requirement, including production levels to be achieved under the technical ecological, social and economic constraints.
- ✦ The production program, range and volume of products to be produced depend on the market requirements, proposed marketing strategy and the availability of resources.
- ✦ A production program should define the levels of output to be achieved during specified periods related to the sales forecast.

- 
- ⌘ Full production level may not be possible during initial production operation owing to various technological, production and commercial difficulties in addition to marketing bottlenecks.

4.4.2 Plant capacity

- Plant capacity (also referred to as production capacity) refers to the volume or number of units that can be manufactured during a given period.
- Plant capacity may be defined in two ways: *feasible normal capacity* and *nominal maximum capacity*.

A. *Feasible normal capacity* refers to the capacity attainable under normal working conditions. This may be established on the basis of the installed capacity, technical conditions of the plant, normal stoppages, and downtime for maintenance, holidays, and shift patterns.

B. The *nominal maximum capacity* is the capacity which is technically attainable and this often corresponds to the installed capacity guaranteed by the supplier of the plant.

Plant capacity is influenced by the following factors:

- *Technological requirement*: For many industrial projects, particularly in process type industries, there is a certain minimum economic size determined by the technological factor. For example, a cement plant should have a capacity of at least 300 tons per day in order to use the rotary kiln method; otherwise, it has to employ the vertical shaft method which is suitable for lower capacity.
- *Input constraints* : In developing countries, there may be constraints on the availability of certain inputs. Power supply and foreign exchange basic raw materials may be scarce;

- *Market conditions*: The anticipated market for the product/service has an important bearing on the plant capacity. If the market for the product is likely to be very strong, a plant of higher capacity is preferable. If the market is likely to be uncertain, it might be advantageous to start with a smaller capacity
- ⌘ *Resources of the firm*: both managerial and financial available resource of a firm define a limit on its capacity decision. Obviously, a firm cannot choose a scale of operations beyond its financial resources and managerial capability.

⌘ *Investment cost*

⌘ *Government Policy*

09/23/2026

4.5. Technology and Engineering Studies

4.5.1. Assessment of technology required

- ✦ The primary goals of technology assessment are to determine and evaluate the impacts of different technologies on the society and national economy (cost-benefits analysis, employment and income effects, satisfaction of human needs etc), impacts on the environment (environmental impact assessment) and techno-economic feasibility assessed from the point of view of the enterprise.

4.5.2. Technology Selection

- ✦ The choice of technology is influenced by variety considerations this are: plant capacity, principal inputs, investment outlay and production cost, use by other units, product mix, and latest developments, ease of absorption, ecological and environmental impact.

- ✦ ***Plant capacity:*** Often, there is a close relationship between plant capacity and production technology. To meet a given capacity requirement perhaps only a certain production technology may be viable.
- ✦ ***Principal inputs;*** The choice of technology depends on the principal inputs available for the project. In some cases, the raw materials available influence the technology chosen.
- ✦ ***Investment outlay and production cost:*** *The* effect of alternative technologies on investment outlay and production cost over a period of time should be carefully assessed.
- ✦ ***Use by other units:*** *The* technology adopted must be proven by successful use by other units, preferably in the specific country

- # ***Latest developments:*** The technology adoption must be based on the latest developments in order to ensure that the likelihood of technological obsolescence in the near future at least, is minimized.
- # ***Ease of absorption:*** The ease with which a particular technology can be absorbed can influence the choice of technology.
- # Sometimes a high level technology may be beyond the absorptive capacity of a developing country which may lack trained personnel to handle that technology.

4.5.3. Appropriateness of technology:

- Appropriate technology should be evaluated in terms of the following questions:
 - # Whether the technology utilizes local raw materials?
 - # Whether the technology utilizes local man power?
 - # Whether the goods and services produced provide the basic needs?
 - # Whether the technology protects ecological balance?

4.5.4. Acquiring Technology

- ✦ When technology has to be obtained from some other enterprises, the means of acquisition have to be determined. These can take the form of *technology licensing* and *outright purchase* of technology

A. **Technology licensing**: this gives the licensee (the one who receive the technology) the right to use the patented technology and get related know how on a mutually agreed basis.

B. **Outright purchase (purchase of technology)**:

4.6 Human Resource and Organization

- # The human resources requirement at various levels and during different stages of the project must be defined as well as their availability and cost.
- # Human resource requirement of the project, the availability of personnel and training needs, the cost estimates for wages, salaries other personnel-related expenses are prepared for the financial analysis of the project.

Categories and Functions

- Human resources as required for the *implementation* and *operation* of project need to be defined by categories such as management as supervision personnel and skilled and unskilled workers and By functions such as general management, production management, accounting, purchase etc.

*Timing of Human Resource requirements*

- 1. Pre -production phase:***
- 2. Operational phase:***

Organizational Set-Up/Design

- ✦ Organization is a means by which the operational functions and activities of the enterprises are structured and assigned to organizational units represented by managerial staff, supervisors and workforce, with the objective of coordinating and controlling the performance of the enterprises and the achievement of its business targets.



Human resource requirements of the project will obviously also depend on:

- The management structure,
- organizational layout,
- operating plan and
- other factors related to the financial and commercial features of the project.

- # The organizational structure of an enterprise indicates the delegation of responsibilities to the various functional units of the company and is normally shown in a diagram.
- # Organizations is designed primarily in line with the different functions in the enterprise such as finance, marketing, purchasing, and manufacturing.
- # However, there is no unique organization pattern. It is also possible to base organizational structures on products or productions lines or on geographical areas or markets



The design of the organization usually includes the following steps:

- I. The goals and objectives for the business
- II. The functions and tasks that are necessary to achieve the goals are identified;
- III. The necessary functions are grouped or related;
- IV. The organizational framework or structure is designed;
- V. All key jobs are analyzed designed and described,
- VI. A recruitment and training program is prepared.

- 
- # The organizational planner will then have to consider some of fundamental aspects of optimal organization. These may include:
 - a. The span of control that is the numbers of employees reporting to supervisor.
 - b. The number of organization levels
 - c. The distribution of responsibilities and authority.



Assessment of supply and demand

- ✦ The following factors should be given due consideration when the availability and employment of human resources are analyzed:
- ✦ The general availability of relevant human resource categories in the country and the project region. The supply and demand situation in the project region.

- # Difficulties in the recruitment of key personnel (such as managers, supervisors and skilled labor) can be dealt with in different ways:
 - Recruitment is combined with **intensive training** of key personnel in order to meet quality requirements
 - Foreign expertise is recruited.
- # An attempt is often made to compensate for the lack of experience of local managerial talent through the employment of foreign personnel, either by hiring individual expatriate or by signing management contracts with foreign companies.

Training Plan

- # Since the lack of experienced and skilled personnel can constitute a significant bottleneck for project implementation and operation especially in developing countries, extensive training programmers should be designed
- # Training can be provided at the factory by managerial and technical personnel and others, by specially recruited experts or by expatriate personnel.

- 
- # The timing of training programs is of crucial importance since personnel should be sufficiently trained to be able to take up their positions as and when required.
 - # Training requirements should be defined separately for the preproduction and for the operation phase in order to provide adequately for production and operational training costs.

Cost Estimates: Estimating the total wage and salary costs.

4.7 Financial and Economic Analysis

- ✦ From a financial and economic point of view, investment can be defined as a long term commitment of economic resources made with the objectives of producing and obtaining net gains (exceeding the total initial investment) in the future.

4.7.2 Project Cash Flows

- # Cash flows are basically either receipt of cash (cash inflow) or payments (cash outflows)
- # Typical operational cash flows for a project are shown below

A. Operational cash outflows

- Increase in fixed assets (investment)
- Operating costs (less depreciation)
- Marketing expenses
- Production and distribution losses
- Corporate (income taxes)



B. Operational cash inflows

- # Revenues from selling of fixed assets
- # Sales revenues
- # Other income due to plant operations



The two methods used to measure projects financial worthiness are:

1. Discounting Cash Flow Method

2. Non-discounting Cash Flow Method

1. Non-Discounted cash flow method

A) Payback period

- This is the number of year taken to recover the original (initial) investment from annual cash flows. The lower the payback period the better the project is

2. cash flow discounting in financial evaluation

- # The basic assumption underlying the discounted cash-flow concept is that money has a time value.
- # A sum of money available now is worth more than an equal sum available in the future.
- # This difference can be expressed as a percentage rate indicting the relative change for a given period which, for practical reasons, is usually a year.

(a) Net Present Value (NPV) Method

$$NPV = \sum_{t=1}^n \frac{C_t}{(1+K)^t} - I_0$$

- # Where; C_t = cash flow at the end of period
- # K = required rate of return
- # n = useful life of project
- # I_0 = initial cost of project
- # NPV = present value of cash flow – present value of initial cost

Decision criteria for NPV

- # $NPV > 0$, Accept the project – it maximizes investors wealth
- # $NPV < 0$, Reject the project
- # $NPV = 0$, Indifferent

A firm is considering investing in a project which costs 6,000 Br and has the following cash flows

YR 1 2 3 4

C.F 1500 3000 2000 2500

The cost of capital is 10% and the project has no salvage value. Using the NPV method advise the firm on whether to invest in the project

Decision: Accept the project since $NPV > 0$

YR	CF	PVIF (10%)	P.Vs
1	1500	0.9091	1363.65
2	3000	0.8264	2479.20
3	2000	0.7513	1502.60
4	2500	0.6830	1707.50
		Total P.Vs =	7053.00
		Less project cost	(6000.00)
		NPV =	1053.00

Chapter Five: Project Implementation, Monitoring and Evaluation

5.1. Organization

- The primary formal relationships for organizing are responsibility, authority, and accountability. They enable us to bring together functions, people, and other resources for the purpose of achieving objectives. The framework for organizing these formal relationships is known as the organizational structure.
- It provides the means for clarifying and communicating the lines of responsibility, authority, and accountability.

5.1.1. Line and Staff Organization

- ❑ **Line managers** have the principal responsibility for achieving the goal of the firm and are vested with decision making authority.
- ❑ **Staff Managers** primarily serve in an advisory capacity (supportive function).
- ❑ Within the staff department they enjoy administrative power.
- ❑ The traditional form of organization is quite appropriate for handling established operations

Cont...

- # When **staff specialists** are added to a **line organization** to "advise; "serve;" or "support" the line in some manner, we have a line and staff organization.
- # These specialists contribute to the **effectiveness** and **efficiency** of the organization. Their authority is generally limited to **making recommendation** to the line organization.
- # Sometimes this creates conflict. However, such conflict can be reduced by having **staff specialists** obtain some line experience, which will tend to make them better understand the problems facing the line managers they support.
- # Such functions as human resources management and research and development are typical staff functions.

Cont...

- # Most large organizations belong to this type of organizational structure. These organizations have direct, vertical relationships between different levels and also **specialists** responsible for advising and assisting line managers.
- # Such organizations have both line and staff departments. Staff departments provide **line people with advice and assistance** in specialized areas(for example, quality control advising production department).

Cont...

5.1.2. Divisional Organization

- A divisional organizational structure gives a larger business enterprise the ability to segregate large sections of the company's business into semi-autonomous groups. These groups are mostly self-managed and focused upon a narrow aspect of the company's products or services. As with any organization structure, divisions have both strengths and weaknesses.
- Unlike departments, divisions are more autonomous, each with its own top executive-often a vice president-and typically manage their own hiring, budgeting and advertising.
- Though small businesses rarely use a divisional structure, it can work for such firms as advertising agencies which have dedicated staff and budgets that focus on major clients or industries.

Cont...

5.1.3. Matrix Organization

- ✦ It is a permanent organization designed to achieve specific results by using teams of specialists from different functional areas in the organization.
- ✦ This type of organization is often used when the firm has to be highly responsive to a rapidly changing external environment.
- ✦ In matrix structures, there are functional managers and product (or project or business group) managers. Functional manager are in charge of specialized resources such as production, quality control, inventories, scheduling and marketing. Product or business group managers are in charge of one or more products and are authorized to prepare product strategies or business group strategies and call on the various functional managers for the necessary resources.

Cont...

- # The matrix organizational structure divides authority both by functional area and by project. In a matrix structure, each employee answers to two immediate supervisors: a functional supervisor and a project supervisor. The functional supervisor is charged with overseeing employees in a functional area such as marketing or engineering. Project supervisors manage a specific and often impermanent project.
- # The problem with this structure is the negative effects of dual authority similar to that of project organization. The functional managers may lose some of their authority because product managers are given the budgets to purchase internal resources.
- # In a matrix organization, the product or business group managers and functional managers have somewhat equal power. There is possibility of conflict and frustration but the opportunity for prompt and efficient accomplishment is quite high.

5.2. Project Planning

- Project planning is a rational determination of how to **initiate**, **sustain**, and **terminate** a project. After the initiation stage, the project is planned to an appropriate level of detail.
- The main purpose is to plan **time**, **cost** and **resources** adequately to estimate the **work needed** and to effectively **manage risk** during project execution.
- As with the initiation process group, a failure to adequately plan greatly reduces the project's chances of successfully accomplishing its goal

Project Planning generally consists of:

- Determining how to plan
- Developing the scope statement
- Selecting the planning team

Cont...

- Identifying deliverables and creating the work breakdown structure;
- Identifying the activities needed to complete those deliverables and networking the activities in their logical sequence;
- Estimating the resource requirements for the activities;
- Estimating time and cost for activities;
- Developing the schedule;
- Developing the budget;
- Risk planning;
- Gaining formal approval to begin work.
- Additional processes, such as planning for communications and for scope management, identifying roles and responsibilities, determining what to purchase for the project and holding a kick-off meeting are also generally advisable.

Cont...

- For new product development projects, conceptual design of the operation of the final product may be performed concurrent with the project planning activities, and may help to inform the planning team when identifying deliverables and planning activities.

Why Project Planning?

- One of the objectives of project planning is to completely define all work required so that it will be readily identifiable to each project participant.
- This is a necessity in a project environment because:
- -If the task is well understood prior to being performed, much of the work can be preplanned.
- -If the task is not understood, then during the actual task execution more knowledge is gained, in turn, leads to changes in resource allocations, schedules, and priorities.

Cont...

- # The more uncertain the task, the greater the amount of information that must be processed in order to ensure effective performance.
- # These considerations are important in a project environment because each project can be different from others, requiring a variety of different resources, but having to be performed under time, cost, and performance constraints with little margin for error.
- # There are four basic reasons for project planning:
 - To eliminate or reduce uncertainty
 - To improve efficiency of the operation
 - To obtain a better understanding of the objectives
 - To provide a basis for monitoring and controlling work

Cont...

- # Planning is decision making based upon futurity. It is a continuous process of making entrepreneurial decisions with an eye to the future, and methodically organizing the effort needed to carry out these decisions.
- # Furthermore, systematic planning allows an organization to set goals. The alternative to systematic planning is decision making based on history.
- # This generally results in reactive management leading to crisis management, conflict management, and fire fighting.

5.3 Project Control

- Project Control is a formal process in project management.
- The *PMBOK* defines Project Control with the following statement:

“A project management function that involves comparing actual performance with planned performance and taking appropriate corrective action (or directing others to take this action) that will yield the desired outcome in the project when significant differences exist.”

Project Control involves the regular review of metrics and reports that will identify variances from the project baseline. The variances are determined by comparing the actual performance metrics in the execution phase against the baseline metrics assigned during the Planning Phase.

These variances are incorporated into control processes to evaluate their meaning.

Cont...

- If significant variances are discovered (variances that place the completion of the project in jeopardy) adjustments are made to the project plan.

A significant variance does not explicitly require a change to the project plan but these variances should be reviewed to determine if preventative action is necessary.

Controlling also includes taking preventative action in anticipation of possible problems.

Project Control is important because it may determine the success of the project by the stakeholders.

Project success relates to project cost, completion date, customer expectations, performance, etc.

Cont...

The Project Control Process

- Controlling procedure need to address such questions:
 - Are we on planned track?
 - Are we on expected budget?
 - Are we on planned schedule?
 - Are we delivering what we said we would?
 - Are we meeting quality and performance standards?
 - Are we meeting stakeholder expectations?
 - What have we accomplished?
 - Will the project objectives be met?
 - What deviations/variances exist?
 - What corrective actions are we taking?

Cont...

- ✦ The project control includes project status reporting, schedule control, cost control, quality control and risk control.
- ✦ Controlling process consists of following steps:

Establish A Baseline of Measurement:

- ✦ The baseline of measurement is actually represented by your project plan. This includes your control schedule, project budget, and any design or performance specifications related to project deliverables.
- ✦ The estimates embodied in these documents create the basis from which variance is measured. The success of project measurement and control depends on how accurate estimated baseline is? What if an estimate is wrong? What if an element of your baseline is a poor representation of what's actually achievable?
- ✦ In case of variance it is difficult to know who is responsible for variance, estimator or task performer.

Cont...

- ✦ The baseline is derived from the cost and duration information found in work breakdown structure and scheduling decisions.

Measuring Progress and Performance:

- ✦ Time and Budget are two quantitative measures. Many methods are available to measure these two. Qualitative measures (i.e. technical specification, production function, reliability etc.) are difficult to measure. Tracking time performance is easier than monitoring budgets.
- ✦ Some of valuable and essential control techniques are:
 - Gantt Chart
 - Control Chart
 - Critical Path Method (CPM)
 - Program Evaluation and Review Technique (PERT)
 - Critical Ratios

Cont...

Analyzing the Performance:

- # Almost all the projects need to be guided right throughout in order to receive the required and expected output at the end of the project.
- # Project status report must be generated on every week or every month. Comparisons of actual and expected(planned) must be done for proactive correction.

Variances

- # Variances are deviations from plan. Think of a variance as the difference between what was planned and what actually occurred. There are two types of variances: positive variances and negative variances.
- # Positive variances: Positive variances are deviations from plan that indicate an ahead-of-schedule situation has occurred or that an actual cost was less than a planned cost.

Cont...

- # This is a good news to the project manager, who would rather hear that the project is ahead-of-schedule or under budget.
- # Positive variances bring their own set of problems, which can be as serious as negative variances. Positive variances can allow for rescheduling to bring the project to completion early, under budget or both. Resources can be reallocated from ahead-of- schedule projects to behind-schedule projects.
- # Positive variances can also result from schedule slippage. Consider budget. Being under budget means that not all dollars were expended, this may be the direct result of not having completed work that was scheduled for completion during the period.
- # Negative variances: Negative variances are deviations from the plan that indicate a behind-schedule situation has occurred or that an actual cost was greater than a planned cost.

Cont...

- # Being behind a schedule or over budget is not what the project manager wants to hear. Negative variances, just like positive variances, are not necessarily bad news. For example, you might have over spent because you accomplished more work during the report period than was planned.
- # But, in over spending during this period, you could have accomplished the work at less cost than was originally planned.
- # In most cases, negative time variances affect project completion only if they are associated with critical path activities or if the schedule slippage on non critical path activities exceeds the activity's total float.
- # Negative cost variances can result from uncontrollable factors such as cost increases from suppliers or unexpected equipment malfunctions. Some negative variances can result from inefficiencies or error.

Cont...

Project Control Techniques

- # Gantt Chart: A Gantt chart, commonly used in project management, is one of the most popular and useful ways of showing activities(tasks or events) displayed against time. Typically, tasks are shown on the vertical axis, and the project time span is represented on the horizontal axis.
- # Each task has a corresponding bar that shows the time span required for that task. The bar can be filled in to show the percentage of the task that has been completed.

Cont...

✚ Gantt Chart represents:

- What the various task/ activities are?
- When each activity begins and ends?
- How long each activity is scheduled to last?
- Where activities overlap with other activities, and by how much overlap?
- The start and end date of the whole project?

✚ Control Chart: Control chart is an efficient way of analyzing performance data to evaluate a project process. Control charts, also known as Shewhart charts named after Walter A. Shewhart. Every process has variation. Some variation may be the result of causes which are not normally present in the process. Some variation is simply the result of numerous, ever-present differences in the process.

Cont...

- # The control chart is a graph used to study how a process changes over time. Data are plotted in time order. A control chart always has a central line for the average, an upper line for the upper control limit (for ahead from scheduled) and a lower line for the lower control limit(for behind the schedule).
- # These lines are determined from historical data. By comparing current data to these lines, we can draw conclusions about whether the process variation is consistent(in control) or is unpredictable(out of control, affected by special causes of variation).
- # One goal of using a Control Chart is to achieve and maintain process stability.

Cont...

The control chart is used to:

- Monitor effects of the variables on the difference between target and actual performance of project.
- Analyze and understand process variables
- Determine process capabilities.

Critical Path Method: The Critical Path Method or Critical Path Analysis is a mathematically based algorithm for scheduling a set of project activities. It is an important tool for effective project management. Critical path method commonly used with all form of projects, including construction, software development, research project, product development, engineering and plant maintenance, among the others.

Cont...

- # Any project with the interdependent activities can apply this technique of scheduling.
- # Critical path method is based on mathematical calculations and it is used for scheduling project activities. In the critical path method, the critical activities of a program or a project are identified. These are the activities that have a direct impact on the completion date of the project.

Cont...

Key Steps in Critical Path Method:

1. Identifying the key activities of project
 2. Determining the Sequence of Activities
 3. Drawing Network Diagram
 4. Time estimate for each activity
 5. Identifying the Critical Path: The critical path is the longest path of the network diagram. The activities in the critical path have an effect on the deadline of the project. If an activity of this path is delayed, the project will be delayed.
- Program Evaluation and Review Technique: CPM assumes low uncertainty in scheduled timings. The Program Evaluation and Review Technique(PERT) is a network model that allows for randomness in activity completion times. PERT was developed in the late 1950's for the U.S. Navy's Polaris project.

Cont...

PERT process involves following steps:

- Specifying the task/activities
 - Determining the sequence of activities
 - Constructing Network Diagram
 - Estimate the Time
 - Critical Path
 - Update the PERT chart
- There are three estimation times involved in PERT; Optimistic Time Estimate(O), Most Likely Time Estimate(M), and Pessimistic Time Estimate (P).
- Optimistic Time Estimate: This time assumes that everything will go according to plan and with minimal difficulties.
- Most Likely Time Estimate: This is the time that, in the mind of the functional manager, would most often occur should this effort be reported over and over again.

Cont...

- # Pessimistic Time Estimate: This time assumes that everything will not go according to plan and maximum difficulties will develop.
- # Critical Ratios: Schedule Performance Index(SPI) and Cost Performance Index(CPI) ratios are critical tools in project management.
- # The budget(Cost) and the schedule (Time) are two important considerations in any project, and since projects that overrun budgets and time frames are not viable for organizations. SPI and CPI ratios are generated from Earn Value System.
- # Schedule Performance Index:
- # The ratio of schedule efficiency that indicates the percent of work performed out of the total work scheduled. A number less than 1 indicates that the project is behind schedule.
- # $SPI = EV/PV$

Cont...

- **Cost Performance Index:**

- The ratio of cost efficiency or value earned per unit actual cost. A number less than 1 indicates that the project is spending more money than budgeted.

- $CPI = EV/AC$

- Lastly, compare the schedule and cost performance indices to each other to determine

- The Critical Ratio (CR), $CR = SPI \times CPI$

- Where:

- SPI = Schedule Performance Index

- CPI = Cost Performance Index

- EV = Earned Value (Budgeted Cost of Work Performed)

- PV = Planned Value (Budgeted Cost of Work Scheduled)

- AC = Actual Cost (Actual Cost of Work Performed)

Cont...

5.4. Human Aspects of Project Management

For the successful execution of a project, a satisfactory human relation is must without such a system other systems of project management do not work well.

To achieve satisfactory human relations in the project setting, the project manager must successfully handle problems and challenges related to: Authority, Orientation, Motivation, Group functioning.

Authority: In project management, the project manager whose activities cut across functional lines of command, lacks the desired formal authority.

While the manager has formal control emanating from contracts and agreement:" as far as outside agencies involved in project work are concerned, in his own organization he has to be contacted with split authority, and dual subordination.

Cont...

- His effective authority would stem from his ability to develop rapport with the project personnel, his skill in professional reputation and stature, his skill in communication and persuasion.

Orientation: Most of the managers, working for a project are engineers or technologist when an engineer assumes managerial responsibility, he faces some different type of problems, which he is supposed to:

- Perform the task of planning, organizing, directing and controlling the resources of the firm in the world of uncertainty.
- Adopt a more creative approach to solve non program and unstructured problems.
- Attach greater importance to efficient utilization of resources and resolution of human relation problem. Thus, for achieving the task he must himself be an accomplished engineer turned manager.

Cont...

- ✦ **Motivation:** The project manager works within the boundaries of a socio-technical system. The principal behavioral factor, which he can influence, is the motivation of the project personnel.
- ✦ In order to succeed in motivating project personnel, the project manager must be a perceptive observer of human beings, must have the ability to appreciate the variable needs of human being, must have a skill in several styles of management suitable to different situations.
- ✦ **Group Functioning:** For building an effective group the company must presume a genuinely participative style of management.
- ✦ With the managerial philosophy the project manager can facilitate the development of mutual trust and acceptance, open communication, co-operation and project attitude.

Cont...

- ✦ In this task the needs leadership capabilities, sensitivity to human nature, perceptiveness, concern for welfare of others, maturity and impartial approach. Actually this is difficult and challenging task.

Cont...

5.5. Pre -requisites for Successful Project Implementation

- Project execution has a direct correlation to project progress and stakeholder's expectations but, time and cost over runs of the project are the main hurdles in successful implementation of a project.
- This problem is very common especially in the public sector. Due to such time and cost over runs, projects tend to become uneconomical, resources are not available to support other projects and overall development is adversely affected.
- To minimize time and cost over runs and thereby to improve the prospects of successful completion of projects, following things can be done:
- 1) Adequate formulation of the project

Cont...

2. Sound project organization.

3. Timely availability of funds.

4. Better contract management.

5. Use of the principle of responsibility accounting.

6. Proper planning for implementation.

7. Judicious tendering and procurement of required equipments.

8. Advance actions and effective monitoring.

4.5 Technology and Engineering Study

Assessment of technology required

- The primary goals of technology assessment are to determine and evaluate the impacts of different technologies on the society and national economy (cost-benefits analysis, employment and income effects, satisfaction of human needs etc), impacts on the environment (environmental impact assessment) and techno-economic feasibility assessed from the point of view of the enterprise. To allow the careful assessment of the suitability of the technological alternatives a logical sequence should be followed: Problem identification, technology description and project layout, technology market and alternatives, assessment of

6. Production Program and Plant Capacity

- The production program, range and volume of products to be produced depend on the market requirements, proposed marketing strategy and the availability of resources.
- A production program should define the levels of output to be achieved during specified periods related to the sales forecast.

(2)Delphi Method: This method is used for eliciting the opinions of a group of experts with the help of a mail survey.

The steps involved in this method are:

- (i) A questionnaire is sent to a group of experts by mail and asked to express their views.
- (ii) The responses received from the experts are summarized without disclosing their identity, and sent back to the experts meant to probe further reasons for extreme views expressed in the first round.
- (iii) The process may be continued for one or more rounds till reasonable agreements emerges in the views of the experts.

II. Time Series Projection Methods

(1) Trend Projection Method

Trend projection method involves the following steps:

- (i) Determine the trend of consumption by analyzing past consumption statistics
- (ii) projecting future consumption by extrapolating the trend.

NB. When the trend projection method is used the most commonly employed relationship is the linear relationship.

$$Y = a + bX$$

...Cont'd

Where; $\hat{Y}$ = Trend value

a = Intercept of the relationship

b = Slope of the relationship

X = Independent variable (time)

$$b = \frac{\sum xy - n\bar{X}\bar{Y}}{\sum x^2 - n\bar{X}^2}$$

$$a = \bar{Y} - b\bar{X}$$

Where, $\bar{X}$ = Mean of X

$\bar{Y}$ = Mean of Y

Advantages of the LSM

- (i) It uses all the observations
- (ii) The straight line is derived by an objective, statistical procedure
- (iii) A measure of goodness of fit is available

Disadvantages of LSM

- (i) It is more complex compared to other methods

(2) Exponential Smoothing Method:

- ❖ In this method forecasts are modified in the light of observed errors.

- $F_{t+1} = F_t + \alpha (D_t - F_t), \quad \text{or,}$

- $F_{t+1} = \alpha D_t + (1 - \alpha) F_t$

Where: α = weight factor for the current demand (the smoothing parameter which lies between 0 and 1).

D_t = demand during the present period

F_{t+1} = Forecast for next period t

F_t = Forecast of demand made for the present period

(3) Moving Average Method:

- ❑ In this method, the forecast for the next period is equal to the average of sales for several preceding periods.

(III) Causal Methods:

- ❑ This method assumes that demand is related to some underlying factor(s) in the environment, and that cause-and-effect relationships are at work.

f. Uncertainties in Demand Forecasting

Demand forecasts are subject to error and uncertainty which arise from three principal sources:

- Data about past and present market
- Methods of forecasting
- Environmental change

g. Market Planning. Prepare a marketing plan for the new product. A marketing plan usually has the following components:

- Current market situation
- Opportunity and issue analysis,
- Objectives
- Marketing strategy
- Action program

4. Raw Materials and Supplies Study



Different materials and other inputs required for operating the project should be identified and their availability, supply and method of estimating operating costs should be analyzed.



In this part of the feasibility study the following can be included:-

- i. Identification of the type of raw materials and supplies to be used in the project.
- ii. All requirements of materials and supplies should be identified and specified in the study considering all socio, economic, commercial, financial, and technical factors.
- iii. The source of materials availability, their users and price of inputs are to be analyzed. The interdependencies between projects, material and input requirements and supply of these items should be considered.
 - location of the available resources, area of supply, access to transport, transport costs and alternate usage of such materials need to be collected.

iv. Costs of raw materials and supplies:-

- The costs of materials and other supplies have to be analyzed in detail to determine project economies.
- Estimating annual operating costs for materials and supplies are to be made explaining the price mechanisms and key factors affecting prices.
- Cost estimates may be expressed either as the cost per unit produced or in terms of a certain production level to conduct sensitivity analysis.

5. Location, Site and Environmental Impact Assessment

5.1. Location

- Location analysis has to identify locations suitable for the industrial project under consideration.
- Traditional approach to industrial location focused, on the proximity of raw materials and market place, mainly with the intention of minimizing transport costs. However, the modern view requires consideration of not only commercial, technical and financial factors, but also of the social and environmental impact a project might have.
- A project potentially located in a number of alternative regions (several alternative locations may have to be considered).

- 
-
- The choice of location is influenced by a variety of considerations:
 - Proximity to raw materials and markets
 - Availability of infrastructure
 - Availability of labor (especially for labor-intensive projects)
 - Technological and process characteristics
 - Government policies, and
 - Other factors like climatic conditions, ease in coping with pollution, general living conditions, etc.

...Cont'd

- ❖ In terms of a basic location model, the *optimum location* is one where the total cost viz., transportation cost, production cost and distribution costs is minimized. This generally implies that:
 - (i) Projects based largely on imported material may need to be located at ports or near terminals.
 - (ii) A project producing *perishable products* or *agro-processing industries* are market oriented and it is advantageous to locate such production near the major consumption centers.
 - (iii) A resource-based projects like *a cement plant* or *a steel mill* should be located close to the source of basic material.
 - (iv) Petroleum products and pharmaceutical can be located at source or near consumption centers or even at some intermediation point.

...Cont'd

- As far as financial feasibility of alternative locations is concerned, the following data-as well as related financial risks, should be assessed.
- ✓ Production costs (including environmental protection costs)
 - ✓ Marketing costs
 - ✓ Investment costs
 - ✓ Revenues
 - ✓ Taxes, subsidies, grant and allowance
 - ✓ Net cash flows

5.2 Site Selection

- ❑ Once the location is decided upon, a specific project site alternative should be defined in the feasibility study. This will require evaluation of the characteristics of each site.
- ❑ When selecting sites within the selected location, the following requirements and conditions are to be assessed:-
 - Ecological conditions on site (soil, site hazards, climate etc)
 - Environment impact (restrictions, standards, guidelines)
 - Socio-economic conditions (restrictions, incentives, requirements)

...Cont'd

- Local infrastructure at site location (existing industrial infrastructure, economic and social infrastructure, availability of critical project inputs such as labor and factory supplies)
- Strategic aspects (corporate strategies regarding possible future extension, supply and marketing policies)
- Cost of land
- Site preparation and development requirements and costs.

...Cont'd

- Topography, altitude and climate may be important for a project as well as access to water, electric power, roads and railways transport.
- Recruitment of managerial staff and labor may be a critical factor for the viability of the project. Development of housing, schools, medical and social center is necessary to attract the required staff and labor force.

NB: Plant location and site selection can be undertaken simultaneously.

5.3 Environmental Impact Assessment

- Is designed to develop an understanding of the environmental consequences of newly planned or existing projects and of any project related activities.
- EIA is part of project planning process. Environmental benefits or costs are usually externalities or side effects that affect the society in whole or in part.

6. **Production Program and Plant Capacity**

- The production program, range and volume of products to be produced depend on the market requirements, proposed marketing strategy and the availability of resources.
- A production program should define the levels of output to be achieved during specified periods related to the sales forecast.
- Full production level may not be possible during initial production operation owing to various technological, production and commercial difficulties in addition to marketing bottlenecks. Normally a production and sales target of 40-50 percent of the capacity for the first year is considered reasonable. Picking up gradually, towards third or fourth year full production level can be achieved.



With regards to plant capacity, generally two capacity terms used in relation to level of operation.

1. **A feasible normal capacity (FNC)** – refers to the capacity achievable under normal working conditions considering the technical conditions of the plant, normal stoppages, downtime for maintenance & tool changes, holiday's, shift pattern and management system applied.
2. **A nominal maximum capacity(NMC)**– is the technically feasible capacity that corresponds to the installed capacity as guaranteed by the supplier of the plant.

7. Technology Selection

- ❑ Appropriate technology selection should be made.
- ❑ The advocates of appropriate technology urge that the technology should be evaluated in terms of the following questions:
 - Whether the technology utilizes local raw materials?
 - Whether the technology utilizes local manpower?
 - Whether the technology protects ecological balance?
 - Whether the goods and services produced satisfy to the basic needs?

- While selecting the best technologies for the proposed project, the following factors must also be given due attention:
 - Technological impact on the environment: The technology that we are going to select should not only be the one that *minimizes pollution*, but should also *preserve the natural resources* and *save renewable resources*.
 - Careful evaluation and assessment of **hazardous technologies** and the use of **toxic materials** at different stages of production should be made.

- Introduction of **obsolete technologies** must also be carefully considered. Acquisition of previously discarded and disassembled production plants should be rechecked carefully.

8. Organization and Human Resource

- ❑ A division of the Company into organizational units, in line with the marketing, supply, production and administrative functions is necessary for efficient management of operations and designing a proper organizational structure in accordance with the corporate strategies and policies.
- ❑ The recommended organization will depend on the social environment as well as techno-economic necessities. The organizational set-up depends to a large extent on the industrial, enterprise, strategies, policies and values of those in power in the organization.



□ A design of the organization usually includes the following steps:

1. Goals and objectives of the business are stated
2. Then functions are identified
3. Functions are grouped or related
4. Organizations structure or framework designed
5. All key jobs are analyzed, designed, and described
6. A recruitment and training program prepared.

- ❑ The two reasons for preparing an organization:-
 1. To achieve optimal coordination and control on all project inputs.
 2. To structure the investment and production costs and to determine the costs linked with corresponding organizational units.

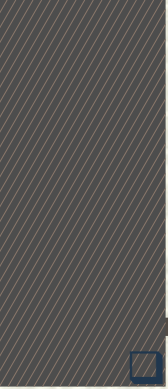
8.1 Organizational Structure

- ❑ Usually the organization structure is designed primarily in line with the different functions. Such as finance, marketing, production and purchasing. However, there is no unique organizational pattern.

Cont.

8.2 Human Resource

- ❑ The successful implementation and operation of industrial projects need different categories of human resources. Example, management, supervisory staff and workers- with sufficient skill and experience.



The following factors should be given due consideration when the availability and employment of human resources are analyzed:-

- i. The general availability of relevant human resource categories in the country and the project region.
- ii. The supply and demand situation in the project region
- iii. Recruitment policy and methods
- iv. Training policy and program

NB. Difficulties in the recruitment of key personnel (such as Managers, supervisors and skilled labor) can be dealt with, in different ways:-

1. Recruitment is combined with intensive training of key personnel in order to meet quality requirements.
2. Foreign expertise is recruited.

9. **Financial and Economic Analysis:**

- ❑ Since reliable cost estimates are fundamental to the appraisal of an investment project it is necessary to check carefully all cost items that could have a significant impact on financial feasibility.
- ❑ Cost estimates cover:-
 - Initial investment cost
 - Cost of production
 - Marketing and distribution costs
 - Plant and equipment replacement costs
 - Working capital requirements and decommissioning at the end of the project life.

9.1 Initial Investment Cost

- ❑ Initial investment costs are the total of fixed assets (fixed asset costs plus pre-production expenditures) and net working capital, with fixed assets constituting the resources required for constructing and equipping an investment project, and net working capital corresponding to the resources needed to operate the project totally or partially.

9.1.1 Pre-production Expenditures:

- In every industrial project certain expenditures are incurred prior to commercial production. They are:-

Preliminary capital – issue expenditures: these are expenditures incurred during the registration and formation of the company. Eg. Legal fees, preparation and issue of a prospectus, ad, public announcement, brokerage commission, etc.

- ii. **Expenditures for preparatory studies:** these includes expenditures for pre-investment studies like opportunity and feasibility and other expenses for planning the project.
- iii. **Other pre-production expenditures:** like
 - Salaries, fringe benefits and social security contributions of personnel engaged during the pre-production period.
 - Travel expenses
 - Preparatory installations, such as work camps, temporary offices and stores.

iv. Cost of trial-runs, start up and commissioning expenditures. These include:

- Fees payable for supervision or start up operations,
- wages, salaries, social security contributions of personnel employed,
- consumption of production materials and supplies, utilities and other incidental start up costs.

9.1.2 Fixed Assets

- ❑ Fixed investment costs should include the following main cost items:
 - Land purchase, site preparation and improvements,
 - Building and civil works
 - Plant machinery and equipment including auxiliary equipment
 - Other assets like industrial property rights and lump sum payments for know-how and patents.

9.1.3 Net working capital

- ❑ Net working capital is defined as current assets minus current liabilities.

9.2 Production Cost

- ❑ It is essential to make realistic forecasts of production and manufacturing costs for a project proposal in order to determine the future viability of the project.
- ❑ Production costs should be determined for the different levels of capacity utilization. The production costs are classified into four major categories. They are:
 - Factory costs
 - Administrative overhead costs
 - Depreciation and cost of financing
 - Operating Cost (the sum of factory and administrative overhead costs).

9.3 Marketing Costs:

- ❑ Marketing costs comprise the costs for all marketing activities and may be divided into direct marketing costs and indirect marketing costs.
 - **Direct marketing costs** – are costs for packaging and storage, sales, product advertisement, transport and distribution costs.
 - **Indirect marketing costs** – are costs related to marketing department. They are salaries for personnel, materials and communication, market research, public relation and promotional activities.

9.4 Cash Flow Statement

- ❑ The cash flow statement shows the movement of cash into and out of the firm and its net impact on the cash balance within the firm.

9.5 Financial Evaluation

- Ranking projects and measuring their profitability have replaced evaluation based on inadequate planning and subjective judgment. Quantitative methods were developed to use in evaluating proposed projects.
- There are two investment evaluation methods (criteria)
 - I. **Discounting criteria** – include NPV, IRR, Benefit-Cost consideration
 - II. **Non-discounting criteria** – include payback period and ARR

I. Discounting Methods (Criteria)

A. Net Present Value Method

- The net present value (NPV) has certain properties that make it a very attractive decision criterion.
- The NPV method is a discounted cash flow method. In this method all net cash inflows are discounted to present value using the *required rate of return* and is then compared with the initial outlay
- If the discounted cash flow exceeds the initial outlay it means the project investigated is attractive since it is expected to earn more than the required rate of return.

$$\text{NPV} = \frac{\text{CF}_1}{(1+r)^1} + \frac{\text{CF}_2}{(1+r)^2} + \frac{\text{CF}_3}{(1+r)^3} + \dots + \frac{\text{CF}_n}{(1+r)^n} - \text{ICO}$$

Where CF= cash inflow per period (year)

r= discount rate

ICO= initial cash outlay

Decision Criteria

- If NPV is greater than zero accept the project
- If NPV is less than zero reject the project

Advantages

1. time value of money is considered
2. It measures the benefits directly
3. It is an objective method of selecting and evaluating projects (By considering cash flows, NPV is not affected by the Co.'s accounting policies, unlike net profit).

Limitations

1. The NPV method does not consider the life of the project. Hence, when mutually exclusive projects with different lives are being considered, the NPV rule is biased in favor of the longer term project.
2. In practice it may be difficult to determine the discount rate. This should relate to the cost of finance, but calculating the costs of the different elements of finance (share capital and loans) is difficult.
3. The NPV is an absolute figure and it does not consider for the size of the project.

Example

To illustrate the calculation of the NPV consider a project which has the following cash flow streams.



Year	Cash Flow
0	\$(1,000,000)
1	200,000
2	200,000
3	300,000
4	300,000
5	350,000

The cost of capital, r , for the firm is 10%. The NPV of proposal is?

...Cont'd

$$\text{NPV} = I_0 - \frac{200,000}{(1.10)^1} - \frac{200,000}{(1.10)^2} - \frac{300,000}{(1.10)^3} - \frac{300,000}{(1.10)^4} - \frac{350,000}{(1.10)^5}$$

= -Br. 5,273

- The NPV represent the net benefit over and above the compensation for the *time* and *risk*

B. Internal Rate of Return (IRR)

- It is another DCF method, which represents the actual rate of return when **profit** and **time value of money** are taken in to account
- It is the rate that **equates** the present value of cash inflow with the present value of cash outflow of an investment.
- It is the discount rate which makes its NPV equal to Zero.

...Cont'd

- ⌘ Hence, the question will be searching for the discounting rate that equates the PV of the investment and cash inflows. That is why IRR is some times called the *internally generated rate of return*. Mathematically, IRR will be obtained when;

$$IO - PV(NCF) = 0$$

- The primary decision rule with IRR is to accept only projects with an IRR greater than the discount rate.

Example

To illustrate the calculation of IRR, consider the cash flows of a project being considered by X Company.

Year	0	1	2	3	4
CF_s	\$(100,000)	30,000	30,000	40,000	45,000

- The IRR is the value of r which satisfies the following equation:

$$\frac{30,000}{(1+r)^1} + \frac{30,000}{(1+r)^2} + \frac{40,000}{(1+r)^3} + \frac{45,000}{(1+r)^4}$$

- The calculation of r involves a process of trial and error. We try different values of r till we find that the right-hand side of the above equation is equal to \$100,000. Let us, to begin with, try $r=15\%$. This makes the right-hand side equal to:

$$\frac{30,000}{(1.15)^1} + \frac{30,000}{(1.15)^2} + \frac{40,000}{(1.15)^3} + \frac{45,000}{(1.15)^4} = 100,802$$

Cont.

- This value is slightly higher than our target value \$100,000. So we must increase the value of r from **15% to 16%**. (In general a higher r lowers and a smaller r increases the right hand side value).
- The right hand side becomes:

$$\frac{30,000}{(1.16)^1} + \frac{30,000}{(1.16)^2} + \frac{40,000}{(1.16)^3} + \frac{45,000}{(1.16)^4} = 98,641$$



Since this value is now less than \$100,000 we conclude that the value of r lies between 15% and 16%. For most of the purposes this indication is sufficient, but if a more refined estimate of r is needed, use the following procedures.

1. Determine the NPV of the two closest rates of return

$$(\text{NPV}/15\%) = - 802$$

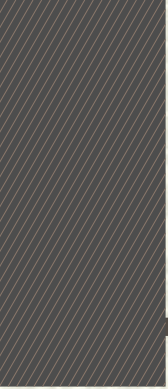
$$(\text{NPV}/16\%) = + 1,359$$

2. Find the sum of the absolute values of the NPVs obtained in step 1

$$802 + 1,359 = 2,161$$

3. Calculate the ratio of the NPV of the smaller discount rate, identified in step 1, to the sum obtained in step 2.

$$802/2,161 = 0.37$$



$$15 + 0.37 = 15.37 \text{ Percent}$$

- The IRR, calculated in this manner, is a very close approximation to the true internal rate of return.
- The decision rule for IRR is as follows:

ACCEPT: if the IRR is greater than the cost of capital

REJECT: if the IRR is less than the cost of capital

...Cont'd

Class Work

A project has a total outlay of Br. 500,000 with the following pattern of cash inflow. Compute the IRR of the project

Year	1	2	3	4	5
CFs	120	120	120	150	150

(in thousands)

...Cont'd

Solution

- i. Find (guess) the starting rate
 $500000/120000=4.167=7\%$
 $500000/150000=3.333=15\%$
- ii. A better approximate (guess) can be found by

$$\frac{0.07+0.15}{2} = 11\% \text{ or } 0.11$$

- iii. Compute the PV of the cash flow at this rate,
 $120,000 \times 2.444 = \$293,280$
 $150,000 \times 0.659 = 98,850$
 $150,000 \times 0.593 = \underline{38,950}$

PVCF	481,080
Io	<u>500,000</u>
NPV	\$-18920

...Cont'd

- # Now, NPV is **negative**, which means the actual IRR is less than 11%.
The following are the next trials at 9% and 10%

9%

$$120,000 \times 2.531 = 303,720$$

$$150,000 \times 0.708 = 106,200$$

$$150,000 \times 0.650 = 97,000$$

$$507,420$$

$$\underline{-500,000}$$

$$\text{NPV} \dots\dots\dots +7,420$$

10%

$$120,000 \times 2.487 = 298,440$$

$$150,000 \times 0.683 = 102,400$$

$$150,000 \times 0.621 = 93,150$$

$$494,040$$

$$\underline{-500,000}$$

$$\text{NPV} \dots\dots\dots -5,960$$

...Cont'd

iv. Now, IRR is above 9% but below 10%; apply interpolation to approximate the true IRR,

$$\frac{7420}{13380} = 0.55$$

Thus, IRR 9.55%

Exercise

- ❑ A project requires a new investment of 20,000 and produces the following cash flows. Compute the IRR of the project

...Cont'd

Year	1	2	3	4	5
CFs	5000	4000	3000	2000	8000

NPV Vs IRR

- For making choice between two projects competing for the funds at the disposal of a concern, the NPV method can give a better choice because it can give idea of ranking of the projects.

Advantages of IRR

- (1) Like NPV, it deals with discounted cash flows and is based on the time value of money.
- (2) The difference between the IRR and the cost of capital indicates the additional return for risk that the project provides.

Disadvantages of IRR

- (1) If there are negative annual cash flows later than year 0, this may lead to more than one possible IRR. In such a case, IRR must be used with great care.
- (2) If a firm has to rank mutually exclusive projects, choosing the project with the highest IRR may result in suboptimal outcome.

C. Benefit-Cost Ratio (Profitability Index)

- The profitability index, also called benefit - cost ratio, is the ratio of the PV of the future net cash inflows to the initial outlay of the project.
- It measures the desirability of the project and evaluates the worth of an investment.

$$PI = \frac{PV(NCF)}{PV(IO)}$$

- In the application of PI, a project is accepted if $PI > 1$, rejected if $PI < 1$ and we remain indifferent if $PI = 1$. It should be noted that when $PI > 1$, NPV is positive; $PI < 1$, NPV is negative and $PI=1$ when NPV is zero.

...Cont'd

Example

After tax cash flows of a small scale tannery project is given below. Find the profitability index if discount rate is assumed to be 12%?

Year	0	1	2	3	4	5
CFs	40,000	15,000	14,000	13,000	12,000	11,000

...Cont'd

Solutions

Year	Cash Flow	Rate = (12%)	PV
1	15,000	0.893	13,395
2	14,000	0.797	11,158
3	13,000	0.712	9,256
4	12,000	0.636	7,636
5	11,000	0.567	6,237
Total			47,678

...Cont'd

Therefore, $PI = \frac{47,678}{40,000} = 1.192$

II. Non-Discounting Criteria

A. Payback Period

- It is one of the most popular and widely used method
- It is defined as the number of years required to recovery the original cash outlay invested in a project
- The payback period can be calculated using the following formula:

$$PBP = \frac{\text{Total Investment}}{\text{Annual Cash Flow}}$$

Example 1

If a project has an investment of Br. 60,000 and annual cash inflow is Br. 15,000 per year for 10 years. Compute the PBP?

$$\text{PBP} = 60,000 / 15,000 = 4 \text{ years}$$

Example 2

If the project cash inflow is not in “annuity form”, **cumulative cash inflow method** may be used to compute that PBP. Assuming an initial investment of Br. 30,000 for the following stream of cash flows and compute the PBP.

...Cont'd

Year	Cash inflows	Cumulative
1	10,000	10,000
2	8,000	18,000
3	12,000	30,000
4	7,000	37,000
5	9,000	46,000
6	3,000	49,000

Cont.

Hence, PBR, is 3 years

Example 3

If the project cumulative cash flow does not exactly match to the investment outlay, but in annuity form of inflow, then compute the PBP in the following way (assuming initial investment of \$10,000)

...Cont'd

Year	Cash inflow	Cumulative
1	4,000	4,000
2	4,000	8,000
3	4,000	12,000
4	4,000	16,000
5	4,000	20,000

$$\text{PBP} = \frac{10,000}{4000} = 2.5 \text{ Years}$$

Example 4

In case the cumulative inflow does not exactly match the amount of investment and the inflow is not in annuity form use the interpolation method (assume an investment outlay of 15,000); compute the PBP.

...Cont'd

Year	Cash in flow	Cumulative
1	3,000	3,000
2	5,000	8,000
3	10,000	18,000
4	2,000	20,000
5	4,000	24,000

$$\text{PBP} = 2 + \frac{(12 \text{ months} \times 7000)}{10,000} = \text{2 years and 8 months or } 2^{8/12} \text{ years}$$

Advantages of Payback Period

- (1) Applying the technique and understanding the concept is easy to understand
- (2) It is quick and simple to use

Disadvantages of Payback Period

- (1) It ignores the cash flows that occur after the payback period. This weakness of the PBP is in stark contrast to both NPV and IRR that consider all cash flows generated by projects.
- (2) It ignores the time value of money. (Of course, this can be overcome by the use of discounted payback in which the cash flows are discounted prior to calculating the payback period).

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Example (Discounted Payback)

Year	Cash flow	PV@10%	Cumulative PV
0	— 1000	— 1000	— 1000
1	600	545	— 455
2	400	330	— 125
3	300	225	+ 100
4	300	205	+ 305

The discounted payback period for **Project A** given an estimate of 2 years, is:

2 years + $125/225 * 12$ months

2 years and 6.7 months

...Cont'd

- ❖ Discounting is particularly useful when cash flows are distant, whereas payback emphasizes the early years of the project, so there is an inherent conflict in the technique.
- ❖ Also, discounting assumes that the cash flows occur at the end of the year whereas payback assumes that they flow **evenly** throughout the year.

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B. Accounting Rate of Return (ARR)

- ❑ This method uses accounting information, as presented by financial statements, to measure profitability of investment.
- ❑ It is sometimes known as *Average Rate of Return* and calculated by dividing the average income after tax by the average investment of project.

$$\text{ARR} = \frac{\text{Average income}}{\text{Average investment}} \times 100 \quad \text{or} \quad \frac{\text{Average Income}}{\text{Total Investment}}$$

Decision Criteria

- Projects which have an ARR equal to or greater than a pre-specified cutoff rate of return — which is usually between 15% and 30% — are accepted otherwise, rejected.

Advantages

1. It simple to calculate
2. It is based on accounting information, which is readily available, and familiar to businessman
3. It considers benefits over the entire life of the project.

Limitations

1. It is based upon accounting profit, not cash flow
2. It does not take into account the time value of money.